Introduction To Green Computing And Its Importance

Foundation of green computing was laid as far back as 1992 with the launching of Energy Star program in the USA. The success of Energy Star motivated other countries to take up the subject for investigation and implementation

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Any technology that aspires to be nature-friendly ought to be green. Recognition of this fact has led to development of green generators, green automobiles, green energy, green chemistry as well as green computing. Green computing is a leap forward for information technology (IT), and more specifically for information and communication technology (ICT). Green computing has emerged as the next wave of ICT.

Motivation for the subject of green computing arose to protect environment against hazards generated at three different states of ICT, namely, information collection (by electronic devices), information processing (through algorithms and storage) and information transportation (through networking and communication).

Carbon dioxide accounts for about eighty per cent of global warming. As a rule of thumb, if world-wide increasing application of ICT is assumed to contribute at least twenty per cent towards carbon dioxide, ICT becomes responsible for sixteen per cent of global warming. This is undoubtedly a cause of concern. As per one research-based estimate, fifty billion devices like computers, mobile phones, sensors, actuators and robots shall connect to the Internet by this year's end, creating even more havoc.

Of course different strategies would be needed to nudge ICT towards green computing, which is necessary to reduce the pollutants generated in collection, processing and transportation of the information. In today's scenario, primary challenge in achieving green computing is to realise energy-efficient devices, energy-efficient processing and energy-efficient networking. Invariably, energy efficiency is required to address reduction in heat dissipation that is basically responsible for emission of carbon dioxide.

In case of electrical, electronic or computer systems, wasteful heat is generated due to thermal vibration of particles in the components. Therefore any brute attempt towards green initiative should have direct or indirect motivation to reduce this ther-